

Phunky.jl

A Multithreaded Phonon Spectral Function Calculator

William L. Wenig Nakib H. Protik

Humboldt-Universität zu Berlin

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Theoretical
Transport
Physics

1 Motivation

- Spectral Transport Properties
- Why Julia?

2 Harmonic Properties

- Dispersion Relation
- Density of States

3 Spectral and Anharmonic Properties

- Phase Space
- Scattering Rates

Outline

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Secular Equation

$$\boxed{W_{\tau\alpha}^s(\bar{q})\omega_s^2(\bar{q}) = \sum_{\tau'\alpha'} D_{\tau\alpha,\tau'\alpha'}(\bar{q})W_{\tau'\alpha'}^s(\bar{q})} \quad (1)$$

$$\text{with } W_{\tau\alpha}^s = \sqrt{M_\tau}\vartheta_{\tau\alpha}^s \quad (2)$$

$$\text{and } D_{\tau\alpha,\tau'\alpha'}(\bar{q}) = \sqrt{M_\tau M_{\tau'}}^{-1} \sum_{\bar{l}', \tau', \alpha'} G_{0,\tau,\alpha} \exp(-i\bar{q} \cdot \bar{r}_{\bar{l}'}) \quad (3)$$

$\tau, \alpha, s, \bar{l} \equiv$ Atom, Cart., Branch, Unitcell Index
 $\bar{r}_{\bar{l}} \equiv$ Unitcell Position $\bar{l}$
 $\vartheta_{\tau\alpha}(\bar{q}) \equiv$ Equilibrium Displacement Amplitude
 $\omega_s(\bar{q}) \equiv$ Eigenfrequencies
 $G_{0,\tau,\alpha} \equiv$ Interatomic Force Constants
 $\bar{l}', \tau', \alpha'$

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Dispersion

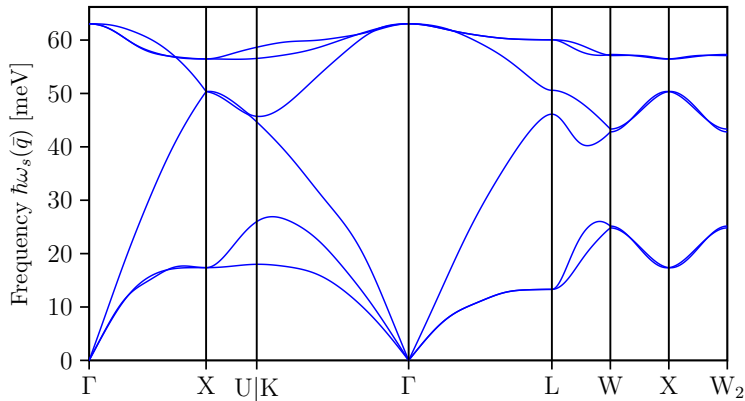
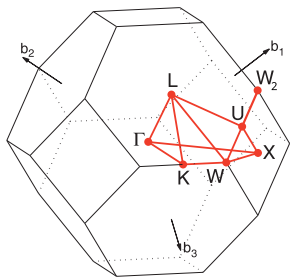


Figure: Harmonic eigenenergies $\omega_s(\bar{q})$ for $\bar{q}$ in cubic Si

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Density of States

$$P^{(2)}(\omega) = \frac{1}{N} \sum_{\vec{q}, s} \delta(\omega - \omega_s(\vec{q}))$$

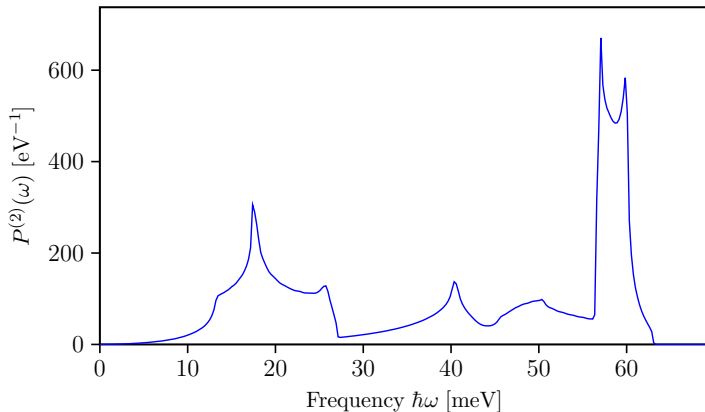


Figure: $P^{(2)}(\omega)$ in cubic Si for a 30^3 $\vec{q}$ -mesh

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3-Phonon Phase Space

$$P^{(3)}(\bar{q}_1, \omega) = \frac{1}{N_2} \sum_{\lambda_2, \lambda_3} \left[\delta(\omega - \omega_{\lambda_2} + \omega_{\lambda_3}) + \frac{1}{2} \delta(\omega - \omega_{\lambda_2} - \omega_{\lambda_3}) \right]$$
$$\lambda = (\bar{q}, s)$$

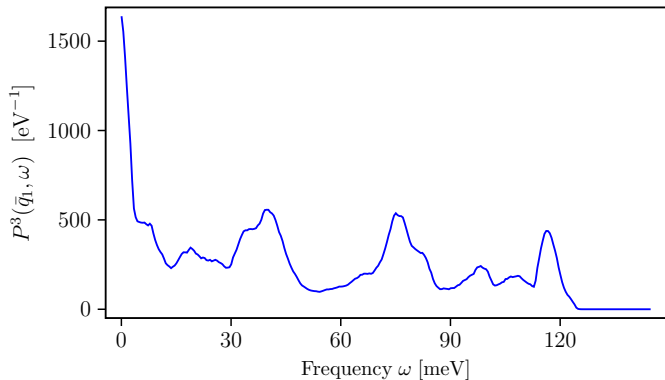


Figure: $P^{(3)}(\bar{q}_1, \omega)$ for $\bar{q}_1 = (0, 0, 1/4)$ in cubic Si on a 24^3 $\bar{q}_2$ -mesh

3-Phonon Phase Space

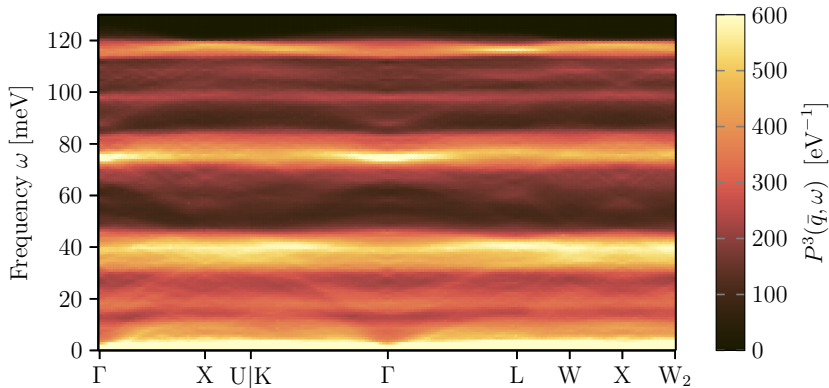
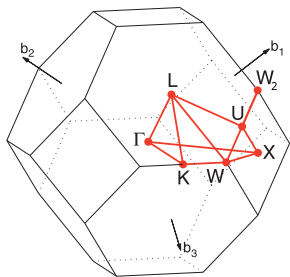


Figure: $P^{(3)}(\bar{q}_1, \omega)$ in cubic Si on a 12^3 $\bar{q}_2$ -mesh

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Scattering Rates

$$\Gamma_{\lambda_1}(\omega) = \frac{1}{\omega_{\lambda_1} N} \left(\Lambda_{\lambda_1}^+ + \frac{1}{2} \Lambda_{\lambda_1}^- \right) \quad (4)$$

$$\Lambda_{\lambda_1}^+ = \sum_{\lambda_2, s_3} \frac{n_{\lambda_2} - n_{\lambda_3}}{\omega_2 \omega_3} \left| V_{\lambda_2 \lambda_3}^+ \right|^2 \delta(\omega - \omega_{\lambda_2} + \omega_{\lambda_3}) \quad (5)$$

$$\Lambda_{\lambda_1}^- = \sum_{\lambda_2, s_3} \frac{n_{\lambda_2} + n_{\lambda_3} + 1}{\omega_2 \omega_3} \left| V_{\lambda_2 \lambda_3}^- \right|^2 \delta(\omega - \omega_{\lambda_2} + \omega_{\lambda_3}) \quad (6)$$

Summary

- Towards Spectral Function with Hilbert Transform
- Towards polar materials with non-analytic contribution
- Enhance summation with tetrahedron Integration
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